

Decision and Risk Analysis

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- ▶ On leaving university, you receive two job offers:
 - ▶ The first is from an **established company** offering a guaranteed salary of **\$150,000**.
 - ▶ The second is from a promising **startup** offering a base salary of **\$100,000** + stock options that will be worth either **\$300,000** or nothing with equal probability.
- ▶ Despite the potential for a large windfall, many people would choose the established company lower but guaranteed amount.
- ▶ This common preference reveals something fundamental about how people make decisions under **risk**.
- ▶ In this part, we will examine the basic frameworks by which economists analyze such decisions under risk, exploring questions such as:
 - ▶ How people evaluate risky choices.
 - ▶ How risk preferences can be modeled.
 - ▶ Where traditional models fail.



\$150,000



\$100,000 +
\$300,000?



Quiz

- ▶ Next ... the concept of **expected value** as a mathematical foundation.
- ▶ Classical framework: **Expected Utility Theory (EUT)**.
 - ▶ Assumes people maximize expected **utility**, not expected value.
 - ▶ Explains **risk aversion** — preference for certain outcomes over risky ones with the same expected value.
- ▶ Yet, real-world behaviour often deviates:
 - ▶ Inconsistent choices between gambles: people tend to overvalue certainty (the “certainty effect”, **Allais Paradox**).
 - ▶ **Framing effects** and **reference dependence**.
 - ▶ **Small-stakes risk aversion** implying unrealistic large-stakes aversion.
- ▶ Understanding EUT strengths and limits prepares us to explore **behavioural alternatives**.

Outline

Introduction

Classical Normative

Descriptive Theories

Prescriptive Theories

Summary and Practical Use

Course Description

Overview of **the state of the art in descriptive and prescriptive theories of decision making under uncertainty** with emphasis on the ways in which **human decisions depart from normative models of rationality**; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.

- ▶ the state of the art in **descriptive** and **prescriptive** theories of decision making **under uncertainty**

Descriptive = how **humans** *actually* decide under uncertainty

Prescriptive = how **agents** *should* decide under uncertainty
(either normatively or using good decision procedures)

Decisions under certainty = the outcome of every alternative is known with certainty and the decision maker simply chooses the best outcome.

Decisions under risk = the decision maker knows the probabilities of each possible state of nature and the goal is to choose the alternative with the highest expected value.

Decisions under uncertainty = the probabilities of outcomes are unknown.

- ▶ **human decisions depart from normative models of rationality ...**

Descriptive

- ▶ The “decision maker” refers to an **actual rational being**.
- ▶ The goal is explanation and prediction of real-world behavior:
 - ▶ How people perceive probabilities and outcomes.
 - ▶ Why they deviate from expected utility (loss aversion, framing effects, etc.).
 - ▶ What cognitive and emotional processes underlie choices (heuristics, attention, learning).
- ▶ The models (Prospect Theory, Regret Theory, heuristics, reinforcement learning) are empirical. They describe what **humans** actually do, not what they ideally should do.

Prescriptive (or Normative)

- ▶ The “decision maker” is an idealized **agent** — a formal construct that could be **human**, machine, or organization.
- ▶ The goal is coherence and optimality, not realism:
 - ▶ What would a perfectly rational agent do, given probabilities and utilities?
 - ▶ What is the optimal policy under uncertainty or limited resources?
 - ▶ How should an AI or policy-maker decide when the cost of being wrong is high?
- ▶ These theories (Expected Utility Theory, Bayesian Decision Theory, MAUT, robust decision frameworks) are normative. They define what should be done to make consistent, optimal, or defensible decisions.

1. Classical Normative (baseline)

Expected Utility Theory (EUT)

Von Neumann & Morgenstern (1944), Savage (1954)

- ▶ Choose the option with the highest expected utility, with a single coherent utility function and subjective probabilities.
- ▶ Assumes independence axiom & Bayesian updating.
- ▶ This is still the canonical *prescriptive* model in economics, game theory, and a lot of AI decision theory.

But descriptively, humans violate it in systematic, not random, ways — which is where modern work comes in.

People overweight small probabilities and underweight large ones, and they evaluate outcomes relative to a reference point (e.g., gains vs. losses) rather than in absolute terms.

These deviations aren't random errors — they follow predictable psychological patterns, which prospect theory captures mathematically.

Example: Expected Utility

► Choice:

- Option A: Receive \$50 for sure.
- Option B: 50% chance of \$100, 50% chance of \$0.

► Both options have the same **expected value**:

$$EV = 0.5 \times 100 + 0.5 \times 0 = 50$$

► Suppose the utility function is $U(x) = \sqrt{x}$ (risk-averse).

$$EU(A) = U(50) = \sqrt{50} = 7.07$$

$$EU(B) = 0.5 \times U(100) + 0.5 \times U(0) = 0.5 \times 10 + 0 = 5$$

► $EU(A) > EU(B)$, so a risk-averse person prefers the sure \$50.

Even with the same expected value, people may prefer certainty due to diminishing marginal utility of money.

2. Descriptive Theories (how people *actually* behave)

- ▶ These try to explain robust deviations from EUT, like loss aversion, over-weighting rare events, ambiguity aversion, etc.
- ▶ Core phenomena: loss aversion, probability distortion, ambiguity aversion.
- ▶ Major families include: **Prospect** Theory, Regret Theory, Ambiguity Aversion, Heuristic Models.

Expected Utility Theory

People maximize expected utility:

$$EU = \sum_i p_i u(x_i)$$

Prospect Theory

People use a **value function** and a **decision weight**:

$$V = \sum_i w(p_i) v(x_i)$$

In EUT, people use true probabilities and a consistent utility curve. In Prospect Theory, they distort both — using psychological weights on probabilities and valuing gains/losses relative to a reference point.

2.1 Prospect (and Cumulative Prospect Theory)

Kahneman & Tversky (1979, 1992)

- ▶ Outcomes evaluated as gains/losses relative to a reference point (e.g., current wealth).
- ▶ Losses hurt more than equivalent gains feel good (loss aversion).
- ▶ People overweigh small probabilities and underweight large ones (probability weighting).
- ▶ Cumulative Prospect Theory (CPT) is the modern version.

Fixes technical problems with the original (like violations of stochastic dominance).

Is widely considered state-of-the-art for descriptive risky choice with known probabilities.

CPT is still the workhorse descriptive model for lab gambles and many field behaviors (lotteries, insurance uptake, etc.).

CPT introduces two key transformations:

1. Value Function

$$v(x) = \begin{cases} x^{0.88}, & x \geq 0, \\ -2.25(-x)^{0.88}, & x < 0 \end{cases}$$

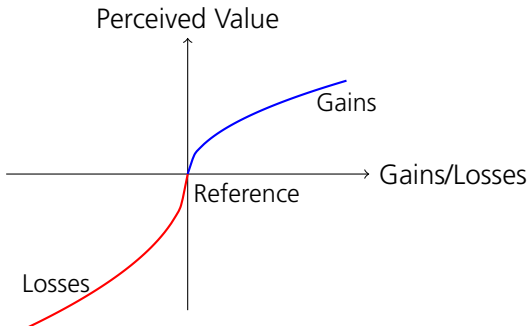
Concave gains (*risk aversion*), convex losses (*risk seeking*).

2. Probability Weighting Function

$$w(p) = \frac{p^{0.61}}{(p^{0.61} + (1 - p)^{0.61})^{1/0.61}}$$

Overweights small probabilities, underweights large ones.

The Value Function

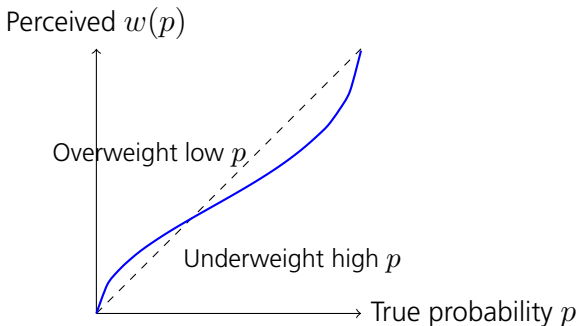


$$v(x) = \begin{cases} x^{\alpha}, & x \geq 0 \\ -\lambda(-x)^{\beta}, & x < 0 \end{cases}$$

where typically $\alpha, \beta \approx 0.88$ and $\lambda \approx 2.25$.

The λ (loss aversion) captures how losses loom larger. The function is concave for gains (risk averse) and convex for losses (risk seeking). Even engineers feel this intuitively when designing with safety margins.

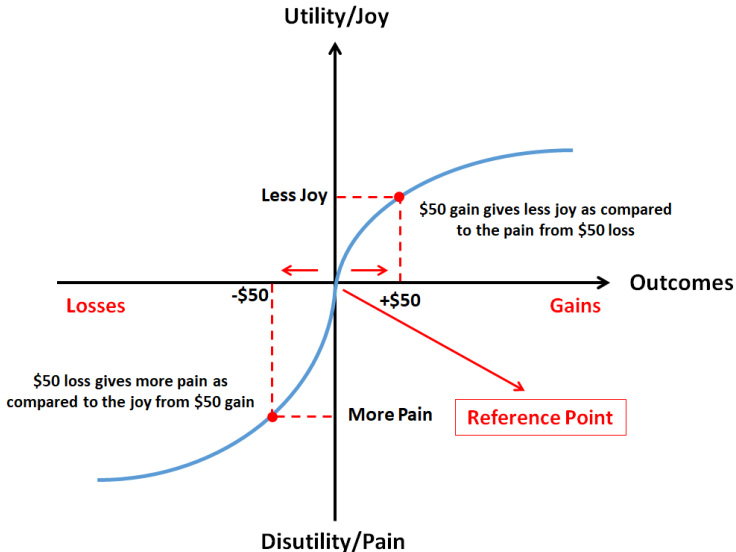
Probability Weighting



$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}}, \quad \gamma \approx 0.61$$

Simple weighting function — low probabilities get exaggerated. “This is why people both buy insurance and lottery tickets.” In engineering, small risks (like rare failures) may dominate design decisions.

Prospect Theory



Example (how CPT explains risky decisions)

CPT explains how people *actually* make risky choices.

Decision Scenario:

- ▶ **Option A (sure gain):** Receive \$100 for sure.
- ▶ **Option B (lottery):** 50% chance to win \$200, otherwise \$0.

Expected Utility Theory (EUT):

$$EU(A) = 100, \quad EU(B) = 0.5 \times 200 + 0.5 \times 0 = 100$$

EUT predicts indifference but people usually prefer the sure \$100.

For gains:

$$v(100) = 100^{0.88} \approx 57, \quad v(200) = 200^{0.88} \approx 104$$

$$w(0.5) \approx 0.42$$

$$V(A) = 57, \quad V(B) = 0.42 \times 104 = 43.7$$

Predicts that most people choose the **sure gain (option A)**.

Now reverse the scenario:

- **Option A:** Lose \$100 for sure.
- **Option B:** 50% chance to lose \$200, 50% to lose nothing.

$$v(-100) = -2.25(100)^{0.88} \approx -128$$

$$v(-200) = -2.25(200)^{0.88} \approx -234 \rightarrow w(0.5) \approx 0.42$$

$$V(A) = -128, \quad V(B) = 0.42 \times (-234) = -98.3$$

Predicts that most people prefer the **risky option (option B)**.

Frame	Option A	Option B	CPT Prediction
Gains	Sure \$100	50% of \$200	Risk averse (A)
Losses	Lose \$100	50% of losing \$200	Risk seeking (B)

CPT captures the reflection effect (people switch from risk aversion in gains to risk seeking in losses, consistent with observed behaviors).

Loss Aversion

- ▶ 50% chance to win \$100, 50% chance to lose \$100.

Most reject it — they require at least a \$200 gain to accept.

$$\text{Losses} \approx 2.25 \times \text{Gains}$$

Engineering analogy

Avoiding a failure (loss) feels more important than achieving a performance boost (gain). Teams prefer to avoid loss of budget or reliability even at the cost of potential innovation gains.

- ▶ **Design decisions:** risk aversion influences safety factors (safety margins chosen higher than necessary due to loss aversion).
- ▶ **Reliability engineering:** perception of rare events affects redundancy design (over designing against rare risks).
- ▶ **Risk communication:** framing probabilities to avoid bias (frame project reliability metrics positively).
- ▶ **Project management:** accounting for human bias in time/cost estimates (planning fallacy in scheduling).

Framing Effects

- ▶ Framing alters perception even though the facts are identical.
- ▶ In engineering communication, clarity of framing affects how stakeholders perceive risk and reliability.

"90% chance of success" sounds better than "10% chance of failure."

"99.9% reliability" vs. "0.1% failure rate."

References

- ▶ Kahneman, D. & Tversky, A. (1979). *Prospect Theory: An Analysis of Decision under Risk*.
- ▶ Kahneman, D. (2011). *Thinking, Fast and Slow*.
- ▶ Tversky, A. & Kahneman, D. (1992). *Advances in Prospect Theory*.

Paper shared on Canvas. Daniel Kahneman won the Nobel Prize in 2002. Read "Thinking, Fast and Slow" for intuition on decision-making under uncertainty.

Homework Assignment

Due: Tue Nov 11 9:35 AM

Read the four scenarios → pick one

Answer the questions (in bullet points or full sentences)

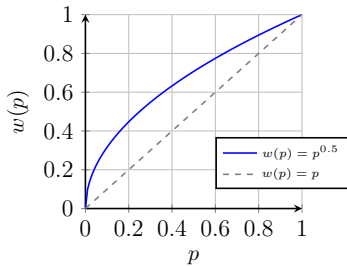
2.2 Rank-Dependent Utility (RDU)

Quiggin (1982) and others

- ▶ Similar in spirit to CPT: instead of using actual probabilities, the decision maker uses transformed “decision weights,” typically overweighting tail events.
- ▶ Captures optimism/pessimism and sensitivity to rank-ordering of outcomes.
- ▶ This is often treated as a unifying framework for probability weighting across models and is still used as a descriptive alternative to EUT.

Weighting Function: e.g., $w(p) = p^{1.7}$

- ▶ The curve shows how actual probabilities p are transformed.
Small probabilities are **overweighted**.
Large probabilities, **underweighted**.
- ▶ Captures optimism/pessimism and attention to rare events.



Probability weighting distorts perception of risk (decision makers act as if low-probability outcomes are even less likely than they are).
Captures lottery and insurance behavior within the same framework.

Example

- ▶ Lottery A: 1% chance to win \$100, 99% chance to win \$0
- ▶ Lottery B: 100% chance to win \$1
- ▶ Expected values are equal (\$1), but many prefer A (the small chance of a big win feels more valuable).
- ▶ RDU explains this by applying a **probability weighting function** (e.g. 1% perceived as 5%).

2.3 Regret Theory

Loomes & Sugden (1982)

- ▶ Assumes choices are driven not just by outcomes but by anticipated regret (or rejoicing) when comparing what you got to what you could have gotten.
- ▶ Explains preference reversals and transitivity violations.
- ▶ Still relevant descriptively in choices where counterfactual comparison is salient (e.g. medical decisions, investments).

Example

- ▶ Choice of two investments:
 - ▶ Option A: 80% chance to win \$100, 20% chance to win \$0
 - ▶ Option B: 100% chance to win \$70
- ▶ By expected value, A (\$80) is better than B (\$70).
- ▶ But if you choose A and get \$0, you will regret not taking the sure \$70.
- ▶ Anticipated regret can make you prefer B (even though A has higher expected value).
- ▶ Explains preference reversals and behavior inconsistent with expected utility.
- ▶ In a different framing (e.g. a gamble game where others are watching), you might choose A to avoid missing out on a possible big win (showing preference reversal, which Regret Theory helps explain).

2.4 Ambiguity Aversion Models

- ▶ Empirically, people treat known risk differently from genuine uncertainty (“ambiguity”), like in Ellsberg-type problems (dislike for unknown probabilities).
- ▶ α -Maxmin / Hurwicz-type criteria: mix best and worst priors with a pessimism parameter α .
- ▶ Smooth Ambiguity Models (Klibanoff, Marinacci, Mukerji, 2005): separate attitudes toward risk from attitudes toward ambiguity by applying a second-order utility to expected utilities over different priors.

These capture robust patterns like “ambiguity aversion in gains but sometimes ambiguity seeking in losses.”

These models sit in a fuzzy space: they’re partly descriptive (they fit human judgment better than Bayesian subjective expected utility in ambiguous environments) and partly prescriptive for ambiguity-averse decision makers.

Example: the Ellsberg Paradox

- ▶ **Urn A:** 50 red and 50 black balls (known probabilities).
- ▶ **Urn B:** 100 balls, red and black, but unknown proportions (ambiguous probabilities).
- ▶ You draw one ball and win \$100 if it's red.

Typical Choice

Most people prefer betting on **Urn A** rather than **Urn B**:

- ▶ Both urns could have the same expected payoff.
- ▶ Yet Urn A feels "safer" because the probabilities are known.

This behavior violates classical **Expected Utility Theory**, which assumes people evaluate outcomes based only on known probabilities.

Maxmin Expected Utility (Gilboa & Schmeidler, 1989)

The decision maker considers a set of possible probability distributions P and evaluates actions by:

$$V(A) = \min_{p \in P} \sum_s p(s) u(A(s))$$

- ▶ Reflects pessimism about the ambiguous situation.
- ▶ In the Ellsberg example, people behave as if they assume the *worst* probability of drawing red from the ambiguous urn.

2.5 Heuristic and Dual-Process Models

- ▶ These don't always give a closed-form utility function, but they do describe psychologically plausible decision processes.
- ▶ Take-the-best: choose based on the first discriminating cue and stop searching.
- ▶ Recognition heuristic: "If I've heard of A but not B, choose A."
- ▶ Fast-and-frugal heuristics (Gigerenzer et al.): people use simple lexicographic or threshold rules that borrow only a few cues and often perform very well in noisy real-world environments.
- ▶ Heuristics-and-biases program (Tversky & Kahneman): availability, representativeness, anchoring. This is more cognitive-process level than formal preference/utility level.

These are now considered mainstream descriptive theories of how decisions happen moment-to-moment, not just which option is picked.

Focus on cognitive process, not formal preference.

People often rely on simple **rules of thumb (heuristics)** to make judgments under uncertainty. Heuristics are efficient but can lead to biases.

Example: Mental Shortcuts (Availability Heuristic)

- ▶ Are there more words that start with the letter **K** or have **K** as the third letter?
- ▶ Most people say: “start with K” because it’s easier to recall words like kite, king, kitchen than words with K as the third letter (asked, maker, token).
- ▶ Rule: Perceived likelihood $\propto$ ease of recall.

Judgments often reflect **memory accessibility**, not actual probabilities.

Dual-Process Models: Two Systems of Thinking

Decision making involves two interacting systems:

- ▶ **System 1:** Fast, automatic, intuitive (uses heuristics).
- ▶ **System 2:** Slow, deliberate, analytical.

Example: The Bat-and-Ball Problem

A bat and a ball cost \$1.10 in total. The bat costs \$1.00 more than the ball. **How much does the ball cost?**

- ▶ Intuitive answer (System 1): \$0.10 — quick but wrong.
- ▶ Analytical answer (System 2): \$0.05 — correct (because $\$1.05 + \$0.05 = \$1.10$).

Dual-process models describe how these two systems interact:

Decision = f (System 1 intuition, System 2 deliberation)

2.6 Reinforcement Learning and Sampling Models

From behavioral econ + computational neuroscience:

- ▶ People often behave as if they're learning action values from experience (reward prediction error style), not solving an optimization problem analytically.
- ▶ Probability weighting can emerge because people sample past outcomes from memory rather than compute exact expected values.

This is now a leading descriptive account of repeated choice under uncertainty and changing environments.

Reinforcement Learning (RL) Models

People learn which actions are good or bad through **trial and error**, updating their expectations based on feedback. They gradually favor actions that yield higher rewards.

Example: Two Snack Machines

- ▶ **Machine A:** Gives a candy bar 70% of the time.
- ▶ **Machine B:** Gives a candy bar 40% of the time.
- ▶ You don't know these probabilities but you learn from experience.

Each time you play, you see the outcome (candy or no candy) and update your beliefs.

Learning Mechanism

At each trial t :

$$Q_{t+1}(a) = Q_t(a) + \alpha[R_t - Q_t(a)]$$

- ▶ $Q_t(a)$: Expected value (estimated reward) of action a
- ▶ R_t : Actual reward (1 = candy, 0 = none)
- ▶ α : Learning rate

Over time, you learn to choose Machine A more often (exploitation), while sometimes testing B (exploration).

Sampling Models

Instead of updating numerical values, people make decisions by **sampling past experiences** from memory. Choices depend on what memories happen to be retrieved.

Example: Remembering Past Outcomes

You've used both machines before. When deciding again:

"Last time I used A, I got candy. The last two times I used B, I didn't."

Even if A isn't objectively better, these recalled samples make A feel like the better option now.

2.7 Reference-Dependent and Goal-Based Models

Modern behavioral models often add:

- ▶ Reference points that move endogenously (e.g. based on expectations).
- ▶ Aspirational targets / satisfying thresholds.
- ▶ Mental accounting (Thaler).

This gives explanations for phenomena like “breaking even” chasing, disposition effect in finance, target-based risk seeking, etc.

These are not always packaged under one grand unified formalism, but reference dependence with dynamic expectations is absolutely considered frontier descriptive work.

Reference-Dependent Models

People evaluate outcomes **relative to a reference point**, not in absolute terms. An outcome feels good or bad depending on whether it is a **gain** or a **loss** compared to that reference.

Example: Salary Satisfaction

Two individuals each earn \$70,000:

- ▶ **Person A:** Expected \$60,000 $\Rightarrow$ feels happy (gain of \$10k).
- ▶ **Person B:** Expected \$80,000 $\Rightarrow$ feels disappointed (loss of \$10k).

Even with identical income, satisfaction differs because each compares to a different reference point.

Utility is defined over **deviations** from a reference point r (Prospect Theory, Kahneman & Tversky, 1979):

$$U(x) = \begin{cases} (x - r)^\alpha & \text{if } x \geq r \text{ (gain)} \\ -\lambda(r - x)^\beta & \text{if } x < r \text{ (loss)} \end{cases}$$

- ▶ $\lambda > 1$: losses loom larger than gains.
- ▶ Explains risk aversion for gains and risk seeking for losses.

Goal-Based Models

People derive utility not only from rewards but from **progress toward goals**. Goals serve as internal reference standards to define success or failure. Goal-based models explain effort patterns, persistence, and post-goal disengagement commonly observed in behavioral and organizational settings.

Example: Performance or Fitness Goals

- ▶ Goal: Run 10 km or sell 100 units.
- ▶ Running 9 km or selling 99 units may feel like *failure*, despite being objectively good.
- ▶ Gain/loss depends on distance to the goal, not absolute value.
- ▶ Utility sharply increases as the agent approaches goal.
- ▶ **Goal-gradient effect:** Motivation rises near goal completion.
- ▶ After achieving the goal, utility flattens ([satiation](#)).

3. Prescriptive / Normative / Decision Analytic

- ▶ Ask how decisions *should* be made under uncertainty.
... given uncertainty in the world and limits in humans,
what's a good way to decide?
- ▶ Two major flavors:
 1. Full-information rationality (Bayesian, utility-maximizing).
 2. Bounded/robust rationality (limited info, cognitive limits).

Example: Medical Treatment Choice

- ▶ A doctor must decide whether to prescribe **Drug A** or **Drug B**.
- ▶ There is uncertainty about each drug's success and side effects.

Drug	P(Success)	P(Severe Side Effect)	Utility Expression
A	0.7	0.10	$0.7U_{\text{success}} - 0.1U_{\text{side effect}}$
B	0.6	0.05	$0.6U_{\text{success}} - 0.05U_{\text{side effect}}$

▶ Prescriptive (Normative) Analysis:

Compute the *expected utility* for each treatment.

Choose the option with the higher expected utility (rational choice).

- ▶ Suppose $U_{\text{success}} = +10$, $U_{\text{side effect}} = -8$.
- ▶ $EU(A) = 0.7(10) - 0.1(8) = 7 - 0.8 = 6.2$
 $EU(B) = 0.6(10) - 0.05(8) = 6 - 0.4 = 5.6$

⇒ **prescriptive recommendation:** choose **Drug A**.

If probabilities or utilities are uncertain, we can use bounded or robust decision rules (e.g., minimax regret or satisficing).

3.1 Full-information Rationality

- ▶ **Bayesian Decision Theory / Subjective Expected Utility:**
maximization with subjective probabilities.
 - ▶ Still the official “gold standard”: assign subjective probabilities, assign utilities, choose max expected utility subject to coherence constraints.
 - ▶ Uses Bayesian updating (Savage, de Finetti).
 - ▶ Backbone of prescriptive analytics, cost-benefit analysis, and rational choice theory.
- ▶ **Multi-Attribute Utility Theory (MAUT):**
 - ▶ Keeney & Raiffa (1976)
 - ▶ Extends EUT to multiple criteria (cost, safety, etc.)

Widely used in policy and risk analysis.

This is considered “state of the practice” in high-stakes domains like energy, defense, pharma trial design, etc.

Example: Investment Choice

- ▶ Suppose you can invest \$1000 in one of two assets.
 - ▶ **Stock A:** 60% chance to double, 40% chance to lose half.
 - ▶ **Stock B:** 80% chance to gain 20%, 20% chance to lose 10%.
- ▶ You know all probabilities and outcomes with certainty.
- ▶ $EU(A) = 0.6(1000 \times 2) + 0.4(1000 \times 0.5) = 1200 + 200 = 1400$
 $EU(B) = 0.8(1000 \times 1.2) + 0.2(1000 \times 0.9) = 960 + 180 = 1140$
⇒ **Decision Rule:** Choose the option with the highest expected utility, **Stock A**.

This is “full-information rationality”—perfect knowledge, no cognitive limits, utility maximization.

Example: Self-Driving Car Navigation

- ▶ A self-driving car must choose between two routes.
 - ▶ **Route A:** 15 min. with 90% probability, 25 minutes with 10%.
 - ▶ **Route B:** Always takes 18 minutes.
- ▶ The car has complete information about travel time probabilities.

Expected Travel Time

$$E[T_A] = 0.9(15) + 0.1(25) = 16 \text{ minutes}$$

$$E[T_B] = 18 \text{ minutes}$$

⇒ **Prescriptive (Normative) Rule:** Choose the route that minimizes expected travel time, **Route A**.

A fully rational agent uses all known probabilities and utilities to make an optimal decision. Humans, by contrast, might prefer Route B to avoid uncertainty (bounded rationality).

3.2 Robust and Bounded Rationality

- ▶ Guard against model misspecification or deep uncertainty.
 - ▶ **Full-information rationality** ("perfect decision maker"): knows probabilities, utilities, and computes optimal choice.
 - ▶ **Bounded rationality** ("smart but limited human"): uses heuristics or shortcuts because of time, attention, or information limits.

Human shortcut: "Highway feels *faster* — I'll just take it unless I see traffic on the app." Simplify the decision instead of computing expected times.
 - ▶ **Robust rationality** ("cautious optimizer"): chooses a decision that performs well enough across a range of uncertain or misspecified conditions, not just the "best guess."
- ▶ **Robust Control, Minimax Regret, Info-Gap:** optimize performance across plausible models.

A cautious driver says: "I don't fully trust the traffic forecast — I'll take the route with the least worst-case delay."
- ▶ Prescriptive in finance, engineering, safety-critical domains.

3.3 Value of Information and Decision Analysis

- ▶ Framework for deciding *what to learn before deciding*.
- ▶ Compute EVPI/EVSI (Expected Value of Information).

The expected improvement in decision quality from obtaining new information before making the decision. In other words:

"How much would I pay to know something before deciding?"

- ▶ Guides data collection, experimentation, and R&D investment.

Example: Value of Information and Decision Analysis

- ▶ You must decide whether to **bring an umbrella**.
- ▶ Probability of rain: 30%.

Option	Weather	P	Utility
Bring	Rain	0.3	+8
Bring	No rain	0.7	-2
Don't	Rain	0.3	-10
Don't	No rain	0.7	+5

Without Information: $EU(\text{Bring}) = +1.0$, $EU(\text{Don't}) = +0.5 \rightarrow \text{Bring}$

With Perfect Information: $EV_{\text{with info}} = 0.3(8) + 0.7(5) = +5.9$

Value of Perfect Information: $VoI = 5.9 - 1.0 = 4.9$

Information is valuable if it changes your decision or improves expected outcomes.

3.4 Multi-Criteria Decision Analysis (MCDA)

- ▶ Structure complex, qualitative decisions.
- ▶ Combine criteria weights and option scores.
- ▶ Includes Analytic Hierarchy Process (AHP).
- ▶ Widely used in policy, engineering, healthcare.

Example: Choosing a Car

- ▶ You want to choose the best car based on multiple criteria:
 - ▶ Price (weight = 0.4)
 - ▶ Fuel efficiency (weight = 0.3)
 - ▶ Comfort / features (weight = 0.3)

Criterion	Weight	Car A	Car B	Car C
Price	0.4	8	6	9
Fuel Efficiency	0.3	6	9	7
Comfort / Features	0.3	9	7	6

Weighted Scores: $A = 7.7, B = 7.2, C = 7.5 \Rightarrow$ **Car A**

MCDA combines multiple conflicting criteria into a single, weighted decision model.

3.5 Satisficing¹ and Bounded Rationality

[1] *Satisficing* is a decision-making strategy or cognitive heuristic that entails searching through the available alternatives until an acceptability threshold is met, without necessarily maximizing any specific objective.

- ▶ Herbert Simon: choose options meeting aspiration levels.
- ▶ Bounded rationality:
 - People have limited time, information, and computational ability — they can't find the perfect choice, so they rely on simplified reasoning or heuristics.
- ▶ Satisficing:
 - Instead of optimizing, we set an aspiration level and choose the first option that meets or exceeds it. ("Good enough," not "best possible.")
- ▶ "Humans aren't maximizing machines — they satisfice within bounds."

Example: Choosing an Apartment

- ▶ People have limited time, information, and mental resources.
- ▶ Instead of maximizing utility, they use heuristics or simple rules.

You are searching for an apartment.

Criterion	Threshold (Aspiration Level)
Rent	$\leq$ \$1500/month
Commute time	$\leq$ 30 minutes
Safety rating	$\geq$ 7/10

Satisficing Rule:

- ▶ Search sequentially through available options.
- ▶ Stop when you find one that meets all thresholds.
- ▶ Accept that option — even if a better one might exist.

3.6 Risk Measures Beyond Expected Utility

- ▶ Tail-risk-sensitive prescriptions:
 - ▶ Expected utility captures average preferences. Risk measures beyond EUT capture how much uncertainty itself matters.
 - ▶ **CVaR minimization:** minimize expected loss in the worst tail.
 - ▶ **Mean–Variance:** simple, regulator-friendly tradeoff.
 - ▶ **Chance Constraints:** limit probability of failure.

Example: Investment Choice

- ▶ You can invest \$100 in one of two options:
 - ▶ **Option A (Safe bond):** \$105 with certainty.
 - ▶ **Option B (Risky stock):** \$140 (50%), \$60 (50%).

Expected Value: $E[A] = 105$, $E[B] = 100 \Rightarrow$ Choose A (under EUT).

Risk measures beyond expected utility capture attitudes toward uncertainty, not just average outcomes.

Alternative Risk Measures:

- ▶ **Variance:** $Var(B)$ is high $\Rightarrow$ risk-averse decision maker prefers A.
- ▶ **Minimax:** Choose option with best worst-case outcome (A = 105, B = 60).
- ▶ **Value-at-Risk (VaR):** "How bad can things get ..." (B = 60, large downside risk).
- ▶ **Mean-Variance Rule:** Maximize $E[X] - \lambda Var(X)$ for risk-return tradeoff.
- ▶ ...

Cheat Sheet

Descriptive Theories (human)

- ▶ Cumulative Prospect Theory
- ▶ Rank-Dependent Utility
- ▶ Regret Theory
- ▶ Ambiguity Models
- ▶ Heuristic & RL models

Prescriptive Theories (agents)

- ▶ Bayesian Decision Theory, MAUT
- ▶ Robust & Ambiguity-Aware methods
- ▶ MCDA, Satisficing
- ▶ Risk-Constrained Optimization

- ▶ Combine behavioral insight with analytical rigor.
- ▶ Use **Cumulative Prospect Theory** and related models for **descriptive** accuracy. (humans)
- ▶ Use **Bayesian / Multi-Attribute Utility Theory + robust methods** for **prescriptive** decisions. (agents)